



AARAVANKIT SHAH

Course : **M.Sc.**, Physics and **B.E.**, Mathematics and Computing , 2028

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CGPA : 9.11



ACADEMIC DETAILS

COURSE	SPECIALIZATION	INSTITUTE/COLLEGE	BOARD/UNIVERSITY	SCORE	YEAR
CLASS XII	Psychology	Prakash College Of Sciences & Commerce	Maharashtra State Board of Secondary and Higher Secondary Education (MSBSHSE)	78 %	2023
CLASS X	Computer Science	Ryan International School	ICSE	95 %	2021

Subjects / Electives

Object Oriented Programming , Nuclear & Particle Physics , Astrophysics & Astronomy , Quantum Information & Computing , Theory Of Relativity , Data Structures & Algorithm , Quantum Mechanics

Technical Proficiency

REST, Git, AutoCAD , Advance MS Excel, Github , Docker , C++ Language , Jupyter notebook , SQLite , NetCDF , Django , CI CD , Postman , Linux , Manim , MATLAB , Flask , API Testing , Python3 , GitHub Actions , Aptitude , Shell Scripting , Active Learning , Java , AIML , Gnuplot

INTERNSHIPS

Research Intern , National Centre For Polar And Oceanic Research

May 2025 - Jul 2025

Conducted a **Study of Long-Term Variability of Ocean Sea Level and Currents in the Bay of Bengal Region** under the guidance of **Dr. Arnab Mukherjee** . Generated and predicted **Wind Patterns & Ocean Currents** for 2008-2021 using a **Linear Model trained on Satellite Data** from 2000-2007 by using *NetCDF* & Python in a terminal based interface .
Conducted a rigorous analysis to **provide reasoning for discrepancies** observed between numerical model predictions and actual satellite derived wind data . Documented & Published the final document to the NCPOR Library . View at ([Link](#)) .

WORK EXPERIENCE

Data Analyst , Birla Institute of Technology and Sciences

Oct 2025 - Nov 2025

Worked on the **Avadaa Energy Ltd.**'s energy data to analyse the factors influencing **Global Horizontal Irradiance** values under **Prof. Pabitra Biswas** , who specialises in *Data Analytics & Statistical Inference* . I *pre-processed the raw sensor data* for further analysis and modelling , ensuring uniformity , compatibility and structural integrity for specialised analytical tools , Developed comprehensive data releases spanning five different timeframes (*1m, 5m, 15m, 30m & 1h*) to enable in-depth , multi-level analysis of changing solar patterns with varying depths . Empirically tested a predictive hypothesis on real-world data to determine the influence of various environmental factors on the Global Horizontal Irradiance (GHI) Values .

Undergraduate Researcher , Birla Institute of Technology and Sciences

Jul 2025 - Dec 2025

Conducting research under *Prof. Sumanta Pasari* to understand the **Socio-Economic Impact of Renewable Energy on Rural India** to help improve **Solar Adaptability** in the country .

- Performed advanced *Multivariate statistical techniques* , including Factor and Cluster Analysis , to classify respondents as Positive , Neutral , and Negative w.r.t. socio-economic impact groups . *Systematically tested* and validated combinations of derived clusters to identify which *economic* factors most significantly influenced each categorised respondent group .
- Developed *targeted , data-driven policy change suggestions and interventions* aimed at shifting the identified Neutral and Negative populations toward the favourable Positive impact categories to improve *India's Solar adaptability* . Paper @[Link](#)

POSITION OF RESPONSIBILITY

Student Faculty Member - Physics Department , BITS Pilani

Jun 2025 - Present

Served as the primary liaison , fostering two-way communication between the Student body and Department Faculty , and translating student feedback into actionable departmental initiatives . My responsibilities included : -

- Serving as the primary liaison between students and the faculty , ensuring transparent feedback and proactively addressing Departmental operational concerns . I facilitated measurable academic improvement by gathering & presenting student feedback on curriculum structure and teaching quality directly to Department heads for review .
- Coordinating key departmental initiatives and large-scale academic events , significantly promoting student engagement and actively assisting faculty with new policy implementation .

PROJECTS

Radiant Player - TUI based Music Player	Mar 2026 - Present
A TUI based , hybrid local and Spotify music player , based on the themes of Brandon Sanderson's Cosmere . It seamlessly mixes local audio files and Spotify tracks in a single unified queue with a beautiful , gemstone -themed dashboard built purely in Bash and Python . It can play Spotify tracks and playlists directly from the terminal without using the desktop app . It can also save the hybrid queues to local playlists and load them later . You can find the souce code at Github Repo .	
Media Tracker - CLI & API Based Media Consumption Tracker	Mar 2026 - Present
An aesthetic personal media tracker designed for Movies /TV Shows /Anime /Manga /Books . This project bridges a Unix CLI toolchain with a glassmorphism -inspired web dashboard to provide a unified , offline experience . Source Code is here .	
Physics Knowledge Base - Physics , API, AI, Learning , Educational	Dec 2025 - May 2026
This project transforms a folder of Markdown notes into a living , rigorous , and self-improving Computational Physics Engine . It goes beyond simple "note-taking " by using AI to structure , link , simulate , and cite complex physical phenomena automatically . <i>Built for the dedicated physicist who values rigour , structure , and computational depth</i> . Github Source Code .	
College Wifi Login - wifi-network	Mar 2025 - Apr 2026
This app lets BITS Pilani students connect to the Wifi-Login Page without having to do the hassle of Inputting credentials every time . This Python script automates login to the BITS Pilani WiFi network . The network portal link automatically redirects users to the login page . The script handles this dynamic redirect process by extracting the required parameters , using POST requests , and verifying internet connectivity and also preventing bot checks by server . Application link & Github Repo .	
Knowledge Grapher - Education	Feb 2026 - Apr 2026
Local exam -prep tool that parses Obsidian -style Markdown into a concept graph , runs PageRank -weighted SM-2 flashcards , and surface weak links . It is a local tool for exam preparation in dense quantitative fields . It parses your Markdown notes (Obsidian -style) , builds a dependency knowledge graph , runs centrality -weighted spaced repetition , and highlights weak links in your understanding . No cloud , no account — your notes and review history stay on your machine . Github Repo .	

CERTIFICATIONS

CERTIFICATION	CERTIFYING AUTHORITY
Postman API Fundamentals Student Expert	Canvas Credentials (Badgr)

AWARDS AND RECOGNITIONS

Math Master Award Ryan International School	Feb 2019
Atal Tinkering Lab's 3D Printing Team Ryan International School	Apr 2019
Super Scientist Award Ryan International School	Feb 2020

CONFERENCES AND WORKSHOPS

Femina Guest Editor
Organized by: Femina , Times Of India Date : Oct 2019
Guest Editor at FEMINA , Times Of India for a day .
Solar Urja Camp
Organized by: IIT Bombay Date : Oct 2018
Created and set the GUINNESS WORLD RECORD for the most artificial lamps (diya's) lit simultaneously at the same time .

SCHOLARSHIPS

Vidyalankar Academic Excellence Scholarship	Jul 2019
Alumni Scholarship	Dec 2024
Alumni Scholarship	Jun 2025
Alumni Scholarship	Dec 2025
BITSAA	May 2026

VOLUNTEER EXPERIENCE

HelpAge India - Role : Volunteer Cause : Social Services	Mar 2018 - Jan 2019
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LANGUAGES KNOWN

English ,Gujrati ,Hindi , Marathi
